



Rincoin 840k S5/b Consensus Change Specification

Extended-Epoch Binary-Halving Transition

Rincoin Community Forge

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1 Status

This is a **Candidate Specification — Discussion Draft** for a Rincoin hard-fork consensus change. Publication does not imply adoption. The specification, reference implementation, tests, activation plan, service readiness, and chain-identification decisions remain under review.

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The key words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** in this document are to be interpreted as described in RFC 2119 and RFC 8174 when, and only when, they appear in all capitals ([Bradner 1997](#); [Leiba 2017](#)).

2 Scope

This specification defines only the maximum permitted block subsidy at and after block height 840,000 under candidate S5/b.

It does not change:

- prior blockchain history or the validity of blocks below height 840,000;
- existing UTXOs or ownership claims;
- RinHash or its parameters;
- Dark Gravity Wave v3;
- transaction format or signature rules;
- transaction-fee accounting;
- any unrelated consensus rule.

Transaction replay, network separation, release deployment, and service migration require separate decisions. They are not solved by this subsidy rule.

3 Definitions

Term	Definition
Activation height H	840000
Phase anchor A	630000
Long reward epoch length L	2100000 blocks
Base unit	10^{-8} RIN; 1 RIN = 100000000 base units

Term	Definition
Base subsidy R	625000000 base units, or 6.25000000 RIN
Post-activation phase index n	$\lfloor (h - A)/L \rfloor$ for height $h \geq H$
Maximum permitted block subsidy	Maximum newly issued amount a coinbase MAY claim at a height, excluding transaction fees
Rounding rule	Integer binary right shift; values below one base unit become zero
First later halving height	2730000
First zero-subsidy height	63630000

The phase anchor is used only to calculate subsidy at and above the activation height. It does not make the extended-epoch rule retroactive and does not alter the validity of any historical block.

4 Pre-Activation Rule

For every block height $h < 840000$, conforming software **MUST** apply the deployed pre-activation subsidy rule without modification. This specification does not restate that rule as new normative text.

The inspected deployed implementation is `GetBlockSubsidy` at commit `b52c87778f800dc5f4e2f59c372bad` ([Rin-coin contributors 2026](#)). The candidate implementation **MUST** demonstrate identical subsidy outputs and block-validity behaviour below activation through regression tests.

5 Activation

The candidate rule activates at **block height 840,000**.

- Height 839,999 **MUST** use the deployed pre-activation rule.
- Height 840,000 **MUST** use the S5/b post-activation calculation with phase index $n = 0$.
- Height 840,001 **MUST** remain in phase index $n = 0$.
- The phase anchor at height 630,000 **MUST NOT** be applied as a new rule to any height below 840,000.

There is no transition window in which either subsidy rule is accepted.

6 Post-Activation Subsidy Rule

For block height $h \geq H$, calculate:

$$n = \left\lfloor \frac{h - A}{L} \right\rfloor.$$

The maximum permitted block subsidy is:

$$\text{Subsidy}(h) = \begin{cases} 625,000,000 \gg n, & 0 \leq n < 30, \\ 0, & n \geq 30. \end{cases}$$

The right shift is an integer operation on a non-negative value. An implementation MUST test $n \geq 30$ before shifting so that it does not perform an out-of-range shift.

This is binary integer halving, nominally a 50% reduction, with floor rounding at base-unit precision. When the preceding value is odd, the effective percentage reduction can differ slightly from 50%.

Consequences:

- height 840,000 returns 625000000 base units;
- height 2,729,999 returns 625000000 base units;
- height 2,730,000 returns 312500000 base units;
- each subsequent 2,100,000-block boundary applies the specified binary integer-halving operation;
- there is no fixed subsidy floor;
- there is no constant terminal phase;
- there is no perpetual tail emission;
- height 63,629,999 returns 1 base unit;
- height 63,630,000 and every later height return zero.

7 Integer Arithmetic

A conforming implementation:

1. MUST represent subsidy in integer base units;
2. MUST calculate the phase index from integer block heights;
3. MUST use binary integer right shift or an exactly equivalent integer-only division by 2^n ;
4. MUST return zero when $n \geq 30$ without performing an out-of-range shift;
5. MUST NOT use floating-point arithmetic in consensus calculation;
6. MUST produce the same result on every supported architecture and build configuration.

Conceptual pseudocode:

```
CAmount GetPost840kS5BSubsidy(int height)
{
    assert(height >= 840000);

    const int64_t phase =
```

```

        ((int64_t)height - 630000) / 2100000;

    if (phase >= 30) {
        return 0;
    }
    return ((CAmount)625000000) >> phase;
}

```

An equivalent implementation MAY be used only if exhaustive phase tests demonstrate identical subsidy outputs and block-validity behaviour.

8 Issuance Accounting

The maximum permitted subsidy MUST be a deterministic function of block height. It MUST NOT depend on:

- actual subsidy claimed by earlier coinbase transactions;
- circulating, spendable, active, lost, or burned supply;
- UTXO state beyond ordinary block validation;
- market price or network hashrate;
- transaction fees.

The total value of coinbase outputs MUST NOT exceed the maximum permitted block subsidy at height h plus the transaction fees collected from that block.

Transaction fees are transfers of existing RIN and are not newly issued supply. A miner MAY claim less than the maximum permitted subsidy. Any underclaimed amount MUST NOT be reissued at a later height.

9 Boundary Vectors

The following vectors are generated from `data/scenario_config.json` by `scripts/simulate_monetary_scenarios.py`. They cover activation, the phase anchor interpretation, early halving boundaries, the final non-zero phase, and the first zero-subsidy block.

Vector	Height	Long-epoch index	Phase anchor	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S5B-001	839999	legacy	n/a	625000000	6.25000000	last pre-activation block
S5B-002	840000	0	630000	625000000	6.25000000	activation block; phase index 0

Vector	Height	Long- epoch index	Phase anchor	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S5B-003	840001	0	630000	625000000	6.25000000	second activation- phase block
S5B-004	2729999	0	630000	625000000	6.25000000	last block of long phase 0
S5B-005	2730000	1	630000	312500000	3.12500000	first block of long phase 1
S5B-006	2730001	1	630000	312500000	3.12500000	second block of long phase 1
S5B-007	4829999	1	630000	312500000	3.12500000	last block of long phase 1
S5B-008	4830000	2	630000	156250000	1.56250000	first block of long phase 2
S5B-009	4830001	2	630000	156250000	1.56250000	second block of long phase 2
S5B-010	6929999	2	630000	156250000	1.56250000	last block before crossing below 1 RIN; last block of long phase 2
S5B-011	6930000	3	630000	78125000	0.78125000	crossing below 1 RIN; first block of long phase 3

Vector	Height	Long- epoch index	Phase anchor	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S5B-012	6930001	3	630000	78125000	0.78125000	second block after crossing below 1 RIN; second block of long phase 3
S5B-013	9029999	3	630000	78125000	0.78125000	last block before crossing below 0.5 RIN; last block of long phase 3
S5B-014	9030000	4	630000	39062500	0.39062500	crossing below 0.5 RIN; first block of long phase 4
S5B-015	9030001	4	630000	39062500	0.39062500	second block after crossing below 0.5 RIN; second block of long phase 4
S5B-016	11129999	4	630000	39062500	0.39062500	last block of long phase 4
S5B-017	11130000	5	630000	19531250	0.19531250	first block of long phase 5
S5B-018	11130001	5	630000	19531250	0.19531250	second block of long phase 5

Vector	Height	Long- epoch index	Phase anchor	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S5B-019	13229999	5	630000	19531250	0.19531250	last block before crossing below 0.1 RIN; last block of long phase 5
S5B-020	13230000	6	630000	9765625	0.09765625	crossing below 0.1 RIN; first block of long phase 6
S5B-021	13230001	6	630000	9765625	0.09765625	second block after crossing below 0.1 RIN; second block of long phase 6
S5B-022	15329999	6	630000	9765625	0.09765625	last block of long phase 6
S5B-023	15330000	7	630000	4882812	0.04882812	first block of long phase 7
S5B-024	15330001	7	630000	4882812	0.04882812	second block of long phase 7
S5B-025	17429999	7	630000	4882812	0.04882812	last block of long phase 7
S5B-026	17430000	8	630000	2441406	0.02441406	first block of long phase 8
S5B-027	17430001	8	630000	2441406	0.02441406	second block of long phase 8

Vector	Height	Long-epoch index	Phase anchor	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S5B-028	19529999	8	630000	2441406	0.02441406	last block of long phase 8
S5B-029	19530000	9	630000	1220703	0.01220703	first block of long phase 9
S5B-030	19530001	9	630000	1220703	0.01220703	second block of long phase 9
S5B-031	21629999	9	630000	1220703	0.01220703	last block of long phase 9
S5B-032	21630000	10	630000	610351	0.00610351	first block of long phase 10
S5B-033	21630001	10	630000	610351	0.00610351	second block of long phase 10
S5B-034	61530000	29	630000	1	0.00000001	first one-base-unit subsidy block
S5B-035	63629999	29	630000	1	0.00000001	final non-zero subsidy block
S5B-036	63630000	30	630000	0	0.00000000	first zero-subsidy block
S5B-037	63630001	30	630000	0	0.00000000	second zero-subsidy block

Machine-readable forms:

- `data/S5B_normative_test_vectors.csv`
- `data/S5B_normative_test_vectors.json`

Artifact	SHA-256
<code>data/S5B_normative_test_vectors.csv</code>	2e970f44cc6590548cfd36701189bad29508a74f53e84ed1685d8e11
<code>data/S5B_normative_test_vectors.json</code>	acf618128185e285c2c20cc50e8d27ed27b4b9e59824854d5be9817

A conforming reference implementation **MUST** match every vector. Before activation readiness is claimed, an independent implementation **SHOULD** generate the same outputs without importing the simulator's subsidy function.

10 Maximum Scheduled Issuance

Under the candidate integer rule:

- maximum scheduled issuance below height 840,000 is 19,687,500.00000000 RIN;
- maximum scheduled issuance at and above height 840,000 is 24,937,499.76900000 RIN;
- total maximum scheduled issuance is 44,624,999.76900000 RIN;
- the first zero-subsidy height is 63,630,000.

The original whitepaper's 168,000,000 RIN statement is a historical upper design bound, not the effective maximum of this rule. No state-dependent mechanism can increase scheduled issuance from 44,624,999.76900000 RIN toward 168,000,000 RIN.

These totals assume every block claims the maximum permitted subsidy. Actual issued and circulating supply **MAY** be lower.

11 Compatibility and Activation Risk

At height 840,000, S5/b permits 6.25000000 RIN, while the unchanged deployed binary-halving rule permits 3.12500000 RIN. S5/b therefore has no immediate miner-revenue shock, but consensus activation divergence is immediate. A block whose coinbase claims more than the legacy amount but no more than the S5/b amount will be valid under S5/b and invalid under unchanged software.

Nodes enforcing different rules can therefore diverge at or after the first block whose coinbase claim distinguishes the rules. Deployment **MUST** be treated as a hard fork. Implementers **MUST** test:

- height 839,999 and 840,000 boundary behaviour;
- heights 2,729,999 and 2,730,000;
- coinbase claims below, equal to, and one base unit above each maximum;
- reorganisation across the activation boundary in both directions;
- initial block download across the boundary;
- block-template subsidy and miner coinbase validation;
- deterministic behaviour on all supported platforms.

12 Chain Identification and Replay

This subsidy change separates block validity but does not, by itself, prevent transaction replay between branches that share transaction formats and signature rules. A block-version marker alone would not provide transaction replay protection.

Before activation readiness is claimed, a separate deployment analysis MUST resolve or explicitly accept:

- P2P and software identity;
- wallet and service chain selection;
- explorer and exchange labelling;
- deposit and withdrawal routing;
- transaction replay exposure;
- whether replay protection requires an additional consensus change.

The chain-identification and replay-risk mechanism (a scheduled coinbase commitment plus a sighash-level `sig_fork_id`) is specified in `technology/consensus-transition.md` §5–6 and `technology/replay-protection-plan.md`; it applies identically across S1, S5/b, and S6/b and is not repeated in this document.

13 Reference Implementation

The candidate implementation is for testing and public review. It is not final, adopted, production-ready, official, or activation-ready.

The public implementation home is the `rincoin-community` GitHub organization. A testing-mode-only candidate implementation is published at `rincoin-community/rincoin-core`, branch `consensus/s5b-testing`, branched from `consensus/s1-testing`, commit `b40c765ff` (CI run, concluded green). This branch refuses to start on mainnet unless `RINCOIN_TESTING_ALLOW_MAINNET=1` is set in the process environment at every invocation; it is not for production use.

Build instructions: `./scripts/prereqs.sh && ./scripts/build.sh` (see that repository’s `README.md` for platform-specific detail).

Test-vector cross-check, executed test results, and the cross-implementation comparison against older Rincoin releases and other independently-developed forks: `verification/S5B_Consensus_Testing_Summary.qmd` and its evidence annex, `verification/S5B/`. Local verification is complete and passing (532 C++ unit tests, 6 fork-scenario functional tests, 159 pre-existing functional tests, all green, including an independent non-C++ re-derivation of this schedule); the full cross-implementation producer × validator matrix and two narrow commitment test-vector cases remain outstanding — see that annex’s own `open-items.md`. The `branch_id` used in the tests matches `technology/consensus-transition.md` §5’s canonical synthetic value (inherited from `consensus/s1-testing`, shared scenario-agnostic infrastructure, not scenario-specific).

14 Non-Normative Rationale

The companion *Rincoin Monetary Policy and Security Scenario Analysis* compares S5/b against S1, the unchanged rule, other extended-epoch schedules, fixed phases, cap-derived decline, Customized Halving study schedules, and perpetual tails. The *Rincoin Monetary*

Review Summary presents the decision path and questions for public review (Rincoin Community Forge 2026a, 2026b).

S5/b allocates the remaining issuance through approximately four-year binary integer halvings, nominally 50%, and avoids an immediate miner-revenue reduction at activation. It is one member of the principal comparison pair in the public-review package; S1 allocates almost the same final issuance through recursive 19/20 reductions, nominally 5%. This rationale is non-normative and does not alter any requirement in this specification.

References

- Bradner, Scott. 1997. *Key Words for Use in RFCs to Indicate Requirement Levels*. RFC 2119. Internet Engineering Task Force. <https://doi.org/10.17487/RFC2119>.
- Leiba, Barry. 2017. *Ambiguity of Uppercase Vs Lowercase in RFC 2119 Key Words*. RFC 8174. Internet Engineering Task Force. <https://doi.org/10.17487/RFC8174>.
- Rin-coin contributors. 2026. *Rincoin Core: Deployed Block-Subsidy Validation*. Version b52c87778f800dc5f4e2f59c372badbc139f933f. Released March 7. <https://github.com/Rin-coin/rincoin/blob/b52c87778f800dc5f4e2f59c372badbc139f933f/src/validation.cpp#L1271-L1281>.
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